

Python Code

```
import pandas as pd
```

```
df = pd.read_csv(' path of a file')
```

```
df.head()
```

```
df.info()
```

```
df.describe(include=' all')
```

```
df.isnull().sum()
```

```
df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda x:  
x.fillna(x.median()))
```

```
df.isnull().sum()
```

```
df.columns=df.columns.str.lower()
```

```
df.columns=df.columns.str.replace(' ','_')
```

```
df.columns=df.columns.str.lower()
```

```
df.columns=df.columns.str.replace(' ','_')
```

```
df=df.rename(columns={'Purchase Amount (USD)': 'purchase_amount'})
```

```
df.columns
```

```
labels=['Young Adult', 'Adult', 'Middle-aged', 'Senior']
```

```
df['age_group']=pd.qcut(df['age'], q=4, labels=labels)
```

```
df[['age', 'age_group']].head(10)
```

```
frequency_mapping = {
```

```
    'Fortnightly': 14,
```

```
    'Weekly': 7,
```

```
    'Monthly': 30,
```

```
    'Quarterly': 90,
```

```
    'Bi-Weekly': 14,
```

```
    'Annually': 365,
```

```
    'Every 3 Months': 90
```

```
}
```

```
df['purchase_frequency_days'] = df['Frequency of Purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days', 'Frequency of Purchases']].head(10)
```

```
print(df.columns.tolist())
```

```
df = df.rename(columns={
```

```
    'Discount Applied': 'discount_applied',
```

```
    'Promo Code Used': 'promo_code_used'
```

```
})
```

```
df[['discount_applied', 'promo_code_used']].head(10)
(df['discount_applied'] == df['promo_code_used']).all()
```

```
df=df.drop('promo_code_used', axis=1)
```

```
df.columns
```

```
pip install psycopg2-binary sqlalchemy
```

```
username = "postgres"
```

```
password = "password"
```

```
host = "local"
```

```
port = "5436"
```

```
database = "Customer_Shopping_Behaviour_Analysis"
```

```
engine = create_engine(
    f"postgresql+psycopg://{username}:{password}@{host}:{port}/{database}"
)
```

```
table_name = "customer"
```

```
df.to_sql(
    table_name,
    engine,
    if_exists="replace",
```

```
        index=False
    )

    print(
        f"Data successfully loaded into table '{table_name}' "
        f"in database '{database}'."
    )

import sys

!{sys.executable} -m pip install "psycpg[binary]"
```